EC24B1039 GAUTAM DEVARAJ <ec24b1039@iiitdm.ac.in>

Fwd: EEG Assignment - II

1 message

Kannadasan K <kannadasankk@iiitdm.ac.in> Thu, Dec 4, 2025 at 12:20 PM
To: ec24b1039@iiitdm.ac.in

Dear Gautam,

An EEG file (.mat format) is attached with this mail for performing the following tasks. The file has 19 channel EEG recordings of a participant which is of duration 10 seconds at a sampling frequency 256 Hz (Dimension: 19 X 2560) . The data is already preprocessed (i.e.) band-pass filtered, eye and muscle artefacts removed.

Plot the EEG signals before and after downsampling.

1. Power Spectral Density (PSD) Analysis

- Compute the Power Spectral Density (PSD) for each EEG channel in the provided preprocessed signal.
- Use appropriate techniques (e.g., Welch's method) to calculate the PSD and plot corresponding figures for each channel as subplots.

2. Mean Power Calculation and Visualization

- Calculate the mean power of each channel across the signal.
- Plot the mean power values for all channels in a clear and labeled graph (line/bar/ of your choice).

3. Downsampling and Reanalysis

- Downsample the EEG signal to 128 Hz while preserving signal integrity.
- Repeat Tasks 1 and 2 on the downsampled signal.

Deliverables:

1. Well-documented Python code implementing the above tasks.
2. Relevant plots and visualizations, saved as image files or embedded in a notebook/report.
3. A brief explanation (200–300 words) of your approach, key observations, and any challenges faced.

--
Regards,
Dr. Kannadasan K Ph.D.,

EEGfile.mat
381K